Dual-Branch Spatial-Wavelet Fusion for Blind Image Deblurring

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***Abstract*—Blind deblurring is a technique that aims to restore a clear image without requiring prior knowledge of the blur kernel. The kernel's variability throughout the image compromises the intricate details and integrity. The spatial domain or the frequency domain are the sole methods employed by the majority of techniques. We introduce WaveFusion-Net, a dual-branch convolutional neural network that is specifically engineered for blind deblurring and employs both. NAFBlocks, SimpleGate, Simplified Channel Attention, and a StripAttention bottleneck are employed by one component to analyze spatial characteristics. The alternative branch applies a Haar discrete wavelet transform (DWT) and refines the low-frequency (LL) subband with WaveletBlocks at two scales. A location-specific sigmoid blend is employed by a trainable CrossBranchFusion module to integrate the branches at those two encoder scales. Training utilizes five distinct loss functions: L1, VGG19 perceptual, FFT, Sobel gradient, and wavelet high-frequency. WaveFusion-Net surpasses Restormer, Uformer, and MPRNet by achieving 29.57 dB / 0.8839 SSIM on RealBlur-J with a mere 9.48M parameters in SHA256-validated, leak-proof divisions. The validation PSNR on the synthetic GoPro and hybrid HIDE benchmarks continues to increase at the conclusion of training within the standard computational limits of 100 and 35 epochs.**

***Index Terms*—wavelet fusion, NAFBlock, CrossBranchFusion, RealBlur-J, blind image deblurring**

I. Introduction

The motion of objects, camera movement, and a lack of focus are the causes of blur in authentic images. Blind deblurring is a method that extracts a clear image from a single blurred observation without the use of a predefined kernel. The inversion is ill-posed as a result of the kernel K being unknown and region-dependent.

Restormer [1] and FFTformer [2] achieved 32.92 and 34.21 dB, respectively, while multiscale convolutional neural networks (CNNs) [3]–[5] forgo direct kernel estimation, achieving 29.08–31.20 dB on GoPro. These baselines were restricted to a single domain.

Spatial CNN processing is executed concurrently with Haar wavelet decomposition by WaveFusion-Net. Localized context is provided by the spatial component, whereas the wavelet component supplies a multi-scale low-frequency (LL) context through WaveletBlocks.

II. Main Contributions

1) Haar wavelet decomposition is fused with spatial CNN processing by WaveFusion-Net via per-scale CrossBranchFusion. The wavelet branch functions as a parallel pipeline that refines the Haar LL band with WaveletBlocks at two scales.
2) We achieved 29.57 dB / 0.8839 Structural Similarity Index (SSIM) on the authentic RealBlur-J benchmark with 2.7 times fewer parameters than Restormer [1], and reported consistent results on GoPro and HIDE within strict computational constraints.
3) We provide a controlled ablation that isolates the wavelet branch (+0.14 dB on GoPro) and the learned gate (+0.04 dB with less than 1% parameter overhead).
4) Using a leakage-free protocol and published SHA256 split hashes, we exhibit reliable generalization across three distinct blur distributions. This enables independent confirmation that no test image influenced the training.

III. Related Work

CNN methods. MPRNet [6] integrates supervised intermediate outputs across stages through Cross-Stage Feature Fusion, achieving 32.66 dB on GoPro and 28.70 dB on RealBlur-J after more than 200 epochs on multi-GPU systems. MIMO-UNet+ [7] implements a multi-input multi-output architecture that follows a coarse-to-fine approach, resulting in scores of 32.45 and 27.63 dB, respectively. In order to achieve a performance of 33.69 dB on GoPro, NAFNet [8] replaces each nonlinear activation with a parameter-free multiplicative gate. Our spatial branch incorporates this block and introduces the parallel frequency pathway, which is absent in NAFNet.

Transformer methods. Uformer [9] implements Locally-enhanced Window attention (32.97 dB GoPro, 29.06 dB RealBlur-J), Restormer [1] applies transposed channel attention (32.92 dB GoPro, 31.22 dB HIDE, 28.96 dB RealBlur-J, 26M parameters), and FFTformer [2] employs FFT-based frequency attention (34.21 dB GoPro), albeit at the expense of spatial locality. All require 200–400 epochs on institutional clusters.

Frequency and wavelet methods. MWCNN [10] utilizes multi-level wavelet decomposition to broaden the receptive field, instead of a branch that is integrated with a spatial pathway. It functions as a singular stream. Wavelets are exclusively employed during the reconstruction phase in SDWNet [11], with no learned gating mechanism between wavelets and spatial attributes. Alam et al. [12] address space-variant blur in saturated, low-light environments, CogSENet [13] combines explicit frequency fusion with blur-conditioned semantic routing, Sun et al. [14] develop a frequency-domain kernel prior, and LoFormer [15] employs localized frequency-domain attention. No one employs per-scale learned gating to integrate a dedicated multi-scale wavelet component with a spatial CNN. Blur statistics are implicitly learned by spatial-exclusive techniques, while frequency-domain approaches sacrifice the spatial locality necessary

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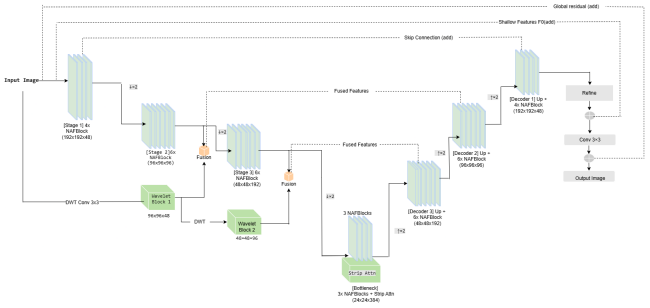


Fig. 1. The comprehensive architecture of WaveFusion-Net consists of a tri-level encoder-decoder pyramid featuring NAFBlocks in the spatial pathway, WaveletBlocks in the concurrent wavelet pathway, CrossBranchFusion at intermediary scales, a StripAttention bottleneck, and additive skip connections (dashed) that carry the fused features to the decoder.

for spatially-varying blur. WaveFusion-Net addresses both of these deficiencies.

IV. Proposed Method

A. Problem Formulation

Let $B \in \mathbb{R}^{3\times H\times W}$ denote an observed blurred image, and $S \in \mathbb{R}^{3\times H\times W}$ denote the associated clear image. The deblurring problem is modeled as

$$B = K * S + \eta \tag{1}$$

where η represents additive noise, $*$ represents convolution, and K represents an unknown spatially-varying blur kernel. We are striving to identify a function $F_\theta : B \to \hat{S}$ such that $\hat{S}$ closely resembles S. The inversion problem is ill-posed, necessitating a learned prior derived from natural image statistics, as K is both unknown and non-uniform.

B. Overall Architecture

WaveFusion-Net is a symmetric encoder-decoder architecture with three encoder levels, a StripAttention bottleneck, and a matching three-level decoder (Fig. 1). It is complemented by a parallel wavelet branch that is connected through two CrossBranchFusion modules. The network learns the correction $\hat{S} - B$ as a result of the forward pass, which includes an image-level residual, $\hat{S} = F_\theta(B) + B$. To ensure compatibility with the three-level $2\times$ pyramid, the input is initially padded with reflection to the nearest multiple of 8, with $h_{\text{pad}} = (8 - H \bmod 8) \bmod 8$ and w_{pad} defined in a corresponding manner. The output is subsequently cropped at the final layer. The model performs 19.57 giga multiply-accumulates (GMACs) at a resolution of 192×192 and comprises 9,484,659 parameters.

C. Spatial Branch

The input is mapped to 48 channels at full resolution using $\text{Conv2d}(3{\to}48, 3\times 3)$, which gives the shallow features $F_0 = W_{\text{intro}}(B_{\text{pad}}) \in \mathbb{R}^{48\times H\times W}$ used by the feature-level residual. Subsequently, four, six, and six NAFBlocks are implemented at 48, 96, and 192 channels by three encoder levels, respectively. This leads to $F_1 \in \mathbb{R}^{48\times H\times W}$, $F_2 \in \mathbb{R}^{96\times H/2\times W/2}$, and $F_3 \in \mathbb{R}^{192\times H/4\times W/4}$. Downsampling, which reduces the resolution by half and doubles the width, employs strided convolutions as $\text{down}_k = \text{Conv2d}(C_k \to 2C_k, 2\times 2, \text{stride } 2)$.

NAFBlock. Each NAFBlock is an activation-free residual block, as per NAFNet [8].

$$\text{NAFBlock}(x) = W_2(\text{SCA}(\text{SG}(\text{DWConv}(W_1(\text{LN}(x)))))) + x \tag{2}$$

In addition to a channel-last LayerNorm (LN), W_1 is defined as $\text{Conv2d}(C{\to}2C, 1\times 1)$, DWConv as $\text{Conv2d}(2C{\to}2C, 3\times 3, \text{groups}{=}2C)$, and W_2 as $\text{Conv2d}(C{\to}C, 1{\times}1)$. SimpleGate (SG) replaces ReLU or GELU with a multiplicative gating mechanism that divides its input into two segments, x_a and x_b, and generates $x_a \odot x_b$. Simplified Channel Attention is defined as $\text{SCA}(x) = x \odot W_{\text{fc}}(\text{GAP}(x))$, where GAP represents global average pooling and W_{fc} represents $\text{Conv2d}(C{\to}C, 1\times 1)$.

D. Wavelet Branch

Motivation. Blur mainly attenuates the high-frequency wavelet subbands, whereas the LL band preserves the coarse scene structure. The branch therefore carries a multi-scale LL representation and fuses it with the spatial features.

An independent shallow layer $W_0 = W_{\text{wav_intro}}(B_{\text{pad}}) \in \mathbb{R}^{48\times H\times W}$ feeds the decomposition and has parameters that are distinct from the spatial branch. The DWT utilizes four static Haar filter banks with a stride of 2 that are stored in non-trainable buffers.

$$F_{\text{ll}} = \tfrac{1}{2}\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}, \qquad F_{\text{lh}} = \tfrac{1}{2}\begin{bmatrix}1 & 1\\ -1 & -1\end{bmatrix} \tag{3}$$

$$F_{\text{hl}} = \tfrac{1}{2}\begin{bmatrix}1 & -1\\ 1 & -1\end{bmatrix}, \qquad F_{\text{hh}} = \tfrac{1}{2}\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix} \tag{4}$$

Each generates a $C \times H/2 \times W/2$ subband, with LL representing coarse structure, LH representing horizontal, HL representing vertical, and HH representing diagonal high-frequency components.

WaveletBlock. Each block refines the LL band with a residual pair of 3×3 convolutions, $\text{LL} + W_{\text{ll}_2}(\text{GELU}(W_{\text{ll}_1}(\text{LL})))$. The three directional subbands are processed by a grouped-convolution path with squeeze-and-excitation gating [16] (Fig. 2(b)), but these outputs are not propagated beyond the block (see Limitations).

Multilevel processing. The DWT is applied at three levels. At the first two, the quartet is refined by WaveletBlock $\mathcal{W}_\ell$ and only $\text{LL}_\ell^{(\text{out})}$ continues, projected by wav_proj_ℓ to the spatial-branch width ($48{\to}96$ and $96{\to}192$, both $1{\times}1$); the first projection is decomposed once more. As inputs, CrossBranchFusion exclusively utilizes the wavelet tensors $w_1 \in \mathbb{R}^{96\times H/2\times W/2}$ and $w_2 \in \mathbb{R}^{192\times H/4\times W/4}$. The third level ($H/8$) is computed but not consumed by any later layer.

E. CrossBranchFusion

The channel quantity is doubled as a consequence of channel concatenation, which is followed by convolution. CrossBranchFusion employs a soft selection that is customized to the specific channel and location, with two distinct scales: 96 channels ($H/2 \times W/2$) and 192 channels ($H/4 \times W/4$). For spatial features F and wavelet features W of matching shape:

$$s = W_s(F), \quad w = W_w(W) \tag{5}$$

$$g = \sigma(W_{\text{gate}}(\text{cat}[s, w])) \tag{6}$$

$$F_{\text{out}} = W_{\text{out}}(g \odot s + (1 - g) \odot w) \tag{7}$$

where all projections are 1×1 convolutions and W_{gate} transforms from $2C$ to C. The gate is intended to favor the spatial branch in sharply defined areas and the wavelet branch in strongly blurred regions. This is accomplished through the use of complementary gates $(g, 1 - g)$, which generate a convex combination that modifies its weighting in response to the input and region during a single forward pass.

F. Bottleneck with StripAttention

The sequence NAFBlock $\to$ StripAttention $\to$ NAFBlock $\to$ NAFBlock is supplied by a third downsampling $F_{\text{bn}} = \text{down3}(F_3) \in \mathbb{R}^{384\times H/8\times W/8}$, all of which operate at 384 channels. In order to

generate directional mappings, StripAttention implements asymmetric depthwise convolutions. DWConv_h (kernel 1×7) and DWConv_v (kernel 7×1) are each configured with groups=C.

$$x_{\text{strip}} = W_{\text{proj}}(\text{cat}[\, x\odot\sigma(\mathrm{DWConv}_h(x)),\ x\odot\sigma(\mathrm{DWConv}_v(x))\,]) \tag{8}$$

At $H/8\times W/8$ each location already covers a large region of the input, so directional attention at this scale suits the global blur direction. Its cost is linear in strip length, against $O((HW/64)^2)$ for full self-attention.

G. Decoder and Skip Connections

The encoder is mirrored in three stages of transposed convolution (2×2, stride 2), with each stage being followed by an identical number of NAFBlocks that correspond to its encoder level (6, 6, 4):

$$D_3 = \mathrm{dec3}(\mathrm{up3}(F_{\text{bottleneck}}) + F_{\text{fused2}}) \tag{9}$$

$$D_2 = \mathrm{dec2}(\mathrm{up2}(D_3) + F_{\text{fused1}}) \tag{10}$$

$$D_1 = \mathrm{dec1}(\mathrm{up1}(D_2) + F_1) \tag{11}$$

Levels 2 and 3 receive the fused CrossBranchFusion features instead of the raw encoder features, so the wavelet-branch (LL) context reaches the reconstruction path only through these two skip connections; the fused features do not enter the next encoder stage. F_1 is employed directly by the finest level, without the use of fusion.

H. Refinement Head and Dual Residual

A refinement head $R = W_{r_2}(\mathrm{GELU}(W_{r_1}(D_1)))$, with both convolutions Conv2d(48→48, 3×3), is combined with the shallow features through a feature-level residual $R_{\text{res}} = R + F_0$. The output projection adds an image-level residual and crops the padding:

$$\hat{S} = \left[W_{\text{outro}}(R_{\text{res}}) + B_{\text{pad}}\right]_{:,:,:H,:W} \tag{12}$$

where $W_{\text{outro}} = \mathrm{Conv2d}(48 \to 3, 3\times3)$, such that the network learns the residual correction $\hat{S} - B$.

V. Loss Function

WaveFusion-Net is trained with a five-component loss covering the pixel, perceptual, frequency, gradient, and wavelet domains.

$$L = 1.0L_1 + 0.1L_{\text{VGG}} + 0.05L_{\text{FFT}} + 0.1L_{\text{Grad}} + 0.02L_{\text{WavHF}} \tag{13}$$

- L_1 (w=1.0): average absolute pixel discrepancy; directly affects the PSNR.
- L_{VGG} (w=0.1): The perceptual loss [17] is guided by the L1 distance between relu1_2, relu2_2, and relu3_4 feature maps of a frozen ImageNet VGG19 [18].
- L_{FFT} (w=0.05): The L1 deviation in the real and imaginary facets of $\mathrm{rfft2}(\cdot)$; blurring primarily distorts the high-frequency phase.
- L_{Grad} (w=0.1): The Sobel gradient L1 error, implemented through grouped convolutions (groups=3), supervises the edges that blur degrades most.
- L_{WavHF} (w=0.02): L1 error on the LH, HL, and HH Haar subbands, which oversees high-frequency intricacies within spatially confined subbands.

VI. Experimental Setup

A. Datasets

Synthetic, hybrid, and authentic blur comprise three benchmarks. GoPro [3] generates motion blur by temporally averaging 240 fps video. This process entails the generation of 2,103 training pairs across 22 sequences and 1,111 test pairs across 11 sequences. HIDE [19] examines blur varieties that are not present in GoPro, with a particular emphasis on human-centered scenarios in which camera tremors and subject movement intersect (6,363 training, 1,329 test pairs). RealBlur-J [20] employs a beam splitter to capture authentic camera shake, which simultaneously reveals both sharp and blurry frames. The dataset contains 3,758 training and 980 test pairs and is devoid of any synthetic elements.

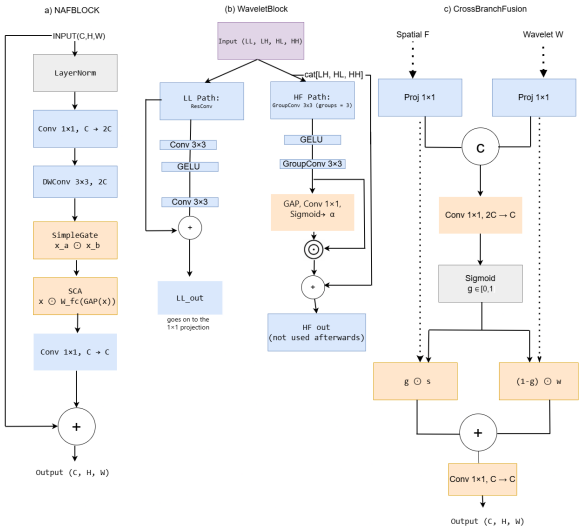


Fig. 2. Component details of (a) NAFBlock, (b) WaveletBlock, and (c) CrossBranchFusion.

B. Evaluation Metrics

The pixel-level reconstruction precision in decibels, which rises as the mean squared error decreases, is denoted by PSNR (peak signal-to-noise ratio, $10\log_{10}(1/\mathrm{MSE})$). SSIM [21] evaluates local brightness, contrast, and configuration within an 11×11 Gaussian window ($\sigma = 1.5$). Both are calculated using normalized RGB in the $[0, 1]$ range at full resolution, without the use of patch tiling.

C. Data Split Protocol and Reproducibility

We establish a deterministic 90/10 train/validation division from each official training dataset in the following manner. 42 is the initial seed of a random number generator. The index array is shuffled in situ. The initial $\lfloor 0.1\cdot N\rfloor$ indices are allocated to the validation set $\mathcal{V}$, while the remainder is assigned to the training set $\mathcal{T}$. The dimensions of the partitions that resulted are presented in Table I.

TABLE I
Dataset split sizes (train / val / test).

Dataset	Train ($\mathcal{T}$)	Val ($\mathcal{V}$)	Test
GoPro	1,893	210	1,111
HIDE	5,727	636	1,329
RealBlur-J	3,383	375	980

At all times, there is no index overlap between $\mathcal{T}$ and $\mathcal{V}$. Checkpoints are evaluated at epoch 1, every 10 epochs, and at the final epoch using the validation PSNR on $\mathcal{V}$. The test dataset is used to evaluate the optimal checkpoint once. No test image had an impact on any training decision.

The SHA256 hash digests of the sorted file path lists were generated as evidence artifacts. Table II contains the initial 12 characters of each. The released repository contains the entire digests.

D. Training Configuration

All experiments were run on Kaggle's 2× NVIDIA Tesla T4 platform (16 GB GDDR6 per GPU) under a 12-hour session limit. The

TABLE II
SHA256 DIGEST PREFIXES FOR EACH SPLIT (FIRST 12 CHARACTERS).

Dataset	Train	Validation	Test
GoPro	`7d659adfcd1f`	`063ab299a04b`	`7018c8b387e9`
HIDE	`e62b33858f1b`	`34a1110dae60`	`e216a016ecbd`
RealBlur-J	`af1027140302`	`8431800b03e0`	`2a98ebbf6075`

model was trained with AdamW [22]; all remaining hyperparameters are listed in Table III.

TABLE III
TRAINING HYPERPARAMETERS.

Hyperparameter	Value
Initial learning rate	2×10^{-4}
Weight decay	1×10^{-4}
LR schedule	CosineAnnealingLR
Minimum LR ($\eta_{\min}$)	1×10^{-7}
Batch size (physical / accumulation / effective)	1 / 2 / 2
Training patch size	192×192
Augmentation	Random crop, h/v flip
Mixed precision	AMP fp16
Gradient clipping	L2 norm ≤ 1.0
Base channels; NAFBlocks (enc/dec/bottleneck)	48; (4,6,6)/(6,6,4)/3

VII. RESULTS AND ANALYSIS

Table IV displays the PSNR and SSIM metrics for the three benchmarks. The baseline metrics are derived from the papers' initial publications.

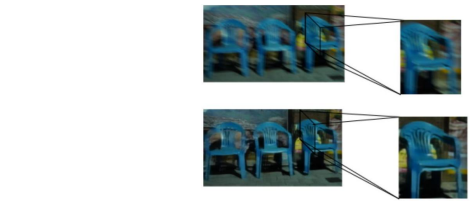
RealBlur-J analysis. With a score of 29.57 dB / 0.8839 SSIM on RealBlur-J, WaveFusion-Net surpasses Restormer [1] by 0.61 dB, Uformer [9] by 0.51 dB, MPRNet [6] by 0.87 dB, and MIMO-UNet+ [7] by 1.94 dB. This is accomplished by utilizing 2.7 times fewer parameters than Restormer.

TABLE IV
BENCHMARK RESULTS (PSNR IN DB AND SSIM); A DASH MARKS A VALUE NOT LISTED.

	GoPro		HIDE		RealBlur-J	
Method	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
DeepDeblur [3]	29.08	0.914	25.73	0.874	27.87	0.827
SRN [4]	30.26	0.934	28.36	0.915	28.56	0.867
DMPHN [5]	31.20	0.940	29.09	0.924	28.42	0.860
MIMO-UNet+ [7]	32.45	0.957	29.99	0.930	27.63	0.837
MPRNet [6]	32.66	0.959	30.96	0.939	28.70	0.873
Restormer [1]	32.92	0.961	31.22	0.942	28.96	0.879
Uformer [9]	32.97	–	30.83	–	29.06	–
FFTformer [2]	34.21	–	31.62	–	–	–
WaveFusion-Net	28.83	0.8754	28.93	0.8937	**29.57**	**0.8839**

GoPro analysis. After 100 epochs, WaveFusion-Net achieved a score of 28.83 dB and 0.8754 SSIM. The validation PSNR continued to increase at the conclusion, rising from 25.89 dB at epoch 1 to 29.70 dB at epoch 100. This increment was +0.06 dB between epochs 80 and 90 and +0.03 dB from 90 to 100. This places WaveFusion-Net below DeepDeblur [3] (29.08 dB) for GoPro. We refrain from making efficiency assertions regarding those benchmarks and attribute the discrepancies in GoPro and HIDE to the training budget rather than the architecture. Restormer [1] and FFTformer [2] are trained for 200–400 epochs on institutional clusters with a significantly larger effective batch size.

HIDE analysis. WaveFusion-Net scores 28.93 dB / 0.8937 SSIM after 35 epochs, the most a 12-hour Kaggle session allows at 18 minutes per epoch. It surpasses SRN [4] by 0.57 dB and DeepDeblur [3] by 3.20 dB, but it is 0.16 dB less than DMPHN [5] (29.09 dB).



(a) RealBlur-J (26.12 dB / 0.7420)



(b) GoPro (27.64 dB / 0.9079)



(c) HIDE (28.90 dB / 0.8814 on this sample)

Fig. 3. Qualitative results. The outputs of WaveFusion-Net are illustrated in Panels (a), (b), and (c), which display per-sample PSNR/SSIM.

TABLE V
ABLATION STUDY ON GOPRO AND REALBLUR-J. HIDE ABLATION WAS OMITTED BECAUSE ONE RUN TAKES ABOUT 11 H AND THE WEEKLY GPU QUOTA IS 30 H; A DASH MARKS A RUN NOT PERFORMED.

		GoPro		RealBlur-J	
Variant	Params	PSNR	SSIM	PSNR	SSIM
Full Model	9.48M	**28.83**	**0.8754**	**29.57**	0.8839
Full Model, seed 7	9.48M	–	–	**29.57**	**0.8841**
Fixed-Fusion (no gate)	9.39M	28.79	0.8748	29.55	**0.8841**
Spatial-Only (no wavelet)	5.31M	28.69	0.8730	29.53	0.8819
No StripAttention	9.18M	28.75	0.8741	**29.57**	0.8836
L1-only loss	9.48M	28.28	0.8652	29.18	0.8775

Qualitative results. Figure 3 illustrates the restored samples from each of the three benchmarks.

Computational complexity. WaveFusion-Net necessitates 19.57 GMACs at a resolution of 192×192 with 9.48 million parameters, as opposed to 36.70 / 26 million for Restormer [1], 29.0 / 20.1 million for Uformer [9], and 15.80 / 4.77 million for NAFNet [8]. This leads to a lower number of GMACs than both transformers, while still achieving a superior RealBlur-J PSNR.

VIII. ABLATION STUDY

Table V displays the single-shot TEST results for the components that were eliminated. The epoch constraints of the respective complete model (GoPro 100, RealBlur-J 70) were adhered to during the training of each component.

Wavelet branch. The removal of the wavelet branch (Spatial-Only) results in a 44% reduction in parameters, with a 0.14 dB decrease in GoPro PSNR and a 0.04 dB decrease in RealBlur-J. The parameter count falls from 9.48 million to 5.31 million, although only 0.46 million of the removed parameters take part in the forward pass (see Limitations). Consequently, the multi-scale LL pathway and its gated fusion provide information that the spatial branch alone does not acquire.

Learned gate. The learned gate was replaced with static concatenation, resulting in a 0.04 dB and 0.02 dB reduction in the PSNR of the GoPro and RealBlur-J, respectively. The gate requires 92,448 parameters, which constitute less than 1% of the total model.

StripAttention. Eliminating StripAttention (301,440 parameters, 3.2% of the model) reduces the GoPro PSNR by 0.08 dB (SSIM 0.8741 compared to 0.8754) while the RealBlur-J PSNR remains at 29.57 dB (SSIM 0.8836 versus 0.8839). This is indistinguishable from the variability between runs on RealBlur-J, and the discrepancy on the GoPro is the result of a single run. Therefore, we consider it to be a negligible advantage at best.

Loss terms. The most substantial impact of the evaluated variants is a PSNR reduction of 0.55 dB on GoPro and 0.39 dB on RealBlur-J, which is achieved by exclusively utilizing L1. The auxiliary loss terms were collectively eliminated, which prevented the isolation of the unique contributions of VGG, FFT, gradient, and wavelet.

Variance and scope. The uniform divisions of Seed 42 were implemented in all variations, as confirmed by SHA256. The primary outcome remained stable across both seeds by executing the complete model on RealBlur-J with an alternative training seed while maintaining a consistent split, as demonstrated by a 29.57 dB / 0.8841 result. The seed-to-seed variation is reflected in the 0.0002 SSIM disparity between the full and Fixed-Fusion models, which is significantly less than the 0.0020 improvement from the wavelet branch. The full-model PSNR is never below that of an ablated variant. All remaining rows contain individual runs.

IX. Limitations

In the trained model, the high-frequency path of the WaveletBlocks and the third wavelet level are computed but not propagated, so they receive no gradient; they hold 3.72 million of the 9.48 million parameters (39%), leaving 5.77 million that influence the output. The gain of the wavelet branch is therefore attributable to the multi-scale LL pathway and its gated fusion rather than to high-frequency modelling, and routing the high-frequency outputs into the fusion is future work. The Haar filters employed in the wavelet component are non-adjustable and static. Adaptable filters, which are constrained to perfect reconstruction, may be capable of accommodating the blur characteristics of a particular dataset. RealBlur-J was the sole dataset evaluated, as the RAW RealBlur-R subset is not available on Kaggle. The model functions as a deterministic regression network; consequently, there has been no investigation of GAN-based adversarial training, which has the potential to enhance perceptual scores at reduced PSNR. It processes each image independently and uses no temporal consistency.

X. Conclusion

We have developed WaveFusion-Net, a dual-branch convolutional neural network that is specifically engineered for blind image deblurring. It combines spatial analysis with a multi-scale Haar-wavelet (LL) pathway by utilizing per-scale CrossBranchFusion modules. CrossBranchFusion utilizes a sigmoid gate to obtain a blend for each channel and spatial location from both branches in order to perform input-adaptive weighting.

WaveFusion-Net achieves 29.57 dB / 0.8839 SSIM on RealBlur-J, surpassing Restormer [1] by +0.61 dB, Uformer [9] by +0.51 dB, MPRNet [6] by +0.87 dB, and MIMO-UNet+ [7] by +1.94 dB. This is accomplished by employing 2.7 times fewer parameters than Restormer on standard hardware. The wavelet branch enhances both datasets, the learned gate provides an increase of 0.02–0.04 dB at a parameter expense of less than 1%, and the multi-term loss contributes an improvement of 0.39–0.55 dB, as demonstrated by the ablation. We provide SHA256 hashes of split file lists for independent verification. The model weights, training notebooks, and split proofs are available at https://github.com/praann07/wavefusion-net-nigercon2026. Future work will route the high-frequency outputs into the fusion and test adaptive wavelet filters.

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